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Bachelor Thesis

Data Visualization for Augmented Reality Head-Up Displays in Personal Protective Equipment

Eingereicht von	Lloyd König
Matrikelnummer	50284
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Betreuer*in	Prof. Dr.-Ing. Georg Freitag
Zweitgutachter*in	Prof. Dr.-Ing. Dietrich Kammer

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1 Introduction

Despite rapid advances and numerous potential applications for augmented reality, it has not yet been widely adopted in industrial settings. One such area of application is in activities where chemical protective clothing is needed. This inherently dangerous work involves significant physical exertion. These activities require constant awareness of the environment and access to information, which is provided by monitoring devices and radio communication in current practice.

The protective equipment restricts the senses and limits mobility, which makes every movement a challenge. A head-up display is able to integrate relevant information into the field of view, thereby enabling hands-free and immediate access, potentially replacing current methods.

However, workers are already pushed to their limits by the equipment and task demands. The HUD must not introduce additional burden—neither in terms of weight nor cognitive load—and it must be possible to utilize the system without interrupting current tasks.

First, the requirements of the intended use case, the limitations imposed by AR technology, and biological constraints are identified. Next, the mechanisms of attention are introduced, including how they can be utilized to design a head-up display capable of providing relevant information without causing distractions. This is followed by a compilation of design options that enable rapid access to and comprehension of information within the aforementioned constraints. Finally, design guidelines are proposed that compile the results of studies on AR and attention management and are illustrated using a potential HUD design.

2 Chemical Protective Clothing and Its Applications

Scenarios where workers are exposed to hazardous materials require chemical protective clothing (CPC). Protection levels range from limited spray protection (Type 6) to fully encapsulating, gas-tight suits (Type 1) with self-contained breathing apparatus (SCBA)[1] ["DGUV Information 212-019"].

Figure 1: Type 1 chemical protection suit 4005/NEO manufactured by isotemp[2]

Confined spaces are defined as areas in which, through their design or site conditions, "inadequate air replacement rates, or the substances [...] found within them or introduced into them, particular hazards exist or may arise that substantially exceed the normal prevailing hazard potential of workplaces" ["DGUV Regel 113-004"]. Tasks that require entering confined spaces are typically cleaning, inspection and/or maintenance of, for example, tanks in chemical plants, rail cars and highway tank trailers and even sewage containers. Traces of the contents may remain, even if their containers are prepared correctly. These residues might—

[1] A self-contained breathing apparatus consists of a respirator connected to a gas cylinder supplying breathable air. The cylinder is fastened to a harness worn on the back.

[2] Vollschutzanzug 4005/NEO · Chemie · isotemp

despite being initially inactive—become dangerous through work procedures (e.g. welding), agitation or chemical reactions ["DGUV Regel 113-004"], hence requiring the highest levels of protection. Other use cases are emergencies, such as confined space rescue, removing chemical spills or other accidents involving unknown, possibly hazardous substances ["OSHA Technical Manual Section VIII"].

These tasks are highly demanding—both physically and mentally—particularly because of the equipment. Since Type 1 suits are not gas-permeable, they compromise the body's ability to regulate temperature, leading to exhaustion and heat fatigue even faster [Holmér, 2006]. The exertion is exacerbated by heavier equipment and heat stress, ultimately reducing work capacity and performance [Smolander et al., 1984]. The maximum working time in a full-body suit with SCBA is therefore limited and might decrease even more depending on oxygen and exhaustion levels.

Further, the material and shape of the suit limit mobility and the field of view to such an extent that even trained personnel may take twice as long to complete tasks when wearing CPC [Murray et al., 2011].

3 Augmented Reality

Augmented reality (AR) is a subcategory of extended reality (XR), alongside mixed reality (MR) and virtual reality (VR). In AR, virtual objects are "superimposed upon or composited with the real world" [Azuma, 1997], in opposition to virtual reality which—by replacing the real environment entirely—creates the deepest sense of immersion. Mixed reality combines features of both concepts, the main distinction being the ability to interact with virtual objects in the real world [Hari Chandana et al., 2023].

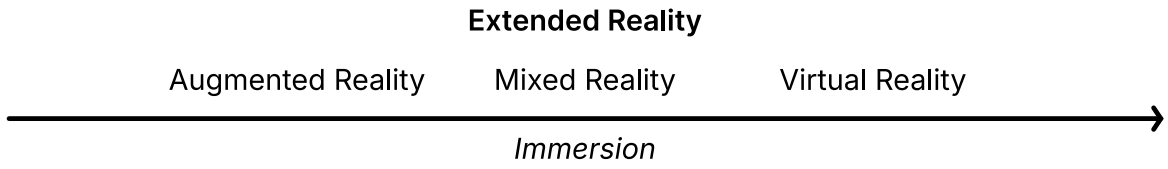

Figure 2: The extended reality spectrum, encompassing augmented, mixed and virtual reality, sorted by degree of immersion, from low to high.

Azuma (1997) defines three requirements for applications to be classified as AR. They combine real and virtual content, they are interactive in real-time, and the virtual objects are registered in 3D space. With the technological advancements since the 2000s these requirements are understood more as guidelines rather than rules.

Head-up displays (HUDs) overlay information onto reality in the user's line of sight, allowing them to access necessary information without having to look away from their primary task. Originally used in military aircraft, today HUDs are installed into consumer vehicles as well. In addition to these integrated HUDs, content can be presented on a head-mounted display. In this case, the display is attached to the head, typically as a helmet or glasses, enabling mobile usage. In contrast to the above definition by Azuma (1997), the content displayed on a HUD is two-dimensional and head-stabilized, rather than spatially anchored, i.e., world-stabilized (see Figure 3). Strictly speaking, a HUD therefore does not meet the criteria for AR. However, other research considers head-stabilized presentation of content as suitable for augmented reality [Feiner et al., 1993, Lu et al., 2020].

G. Lin et al. (2021) also point out that world-stabilized graphics are rarely used in industrial applications, particularly in full-shift scenarios where the HMD would have to be worn continuously. Furthermore, work tasks often require an overview of the situation rather than spatially distributed information. Additionally, with a 2D

HUD, elements are always displayed in the same location, making it easier to find the needed information.

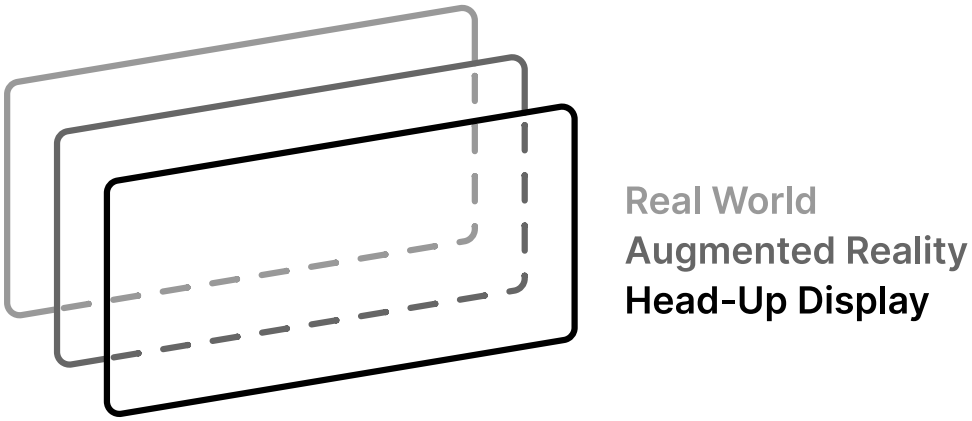

Figure 3: Visualization layers of head-mounted displays: reality, Augmented Reality (3D, world-stabilized) and User Interface (2D, head-stabilized)

3.1 Optical See-Through Displays

There are two approaches to combining virtual content with reality. With video see-through displays, information is superimposed onto a video stream captured by a camera mounted on the device. These headsets fully enclose the head to block external light. In contrast, an optical see-through (OST) system allows the user to view reality directly. Here, the overlay is created by optical combiners blending light from the image source and the environment together.

Since video see-through displays completely block the view of the outside world, it is impossible to see anything in the event of a failure. Given the intended use cases, this represents a major safety issue, especially since the headset cannot be removed while wearing a fully encapsulating suit.

3.1.1 Ambient Light and Contrast Issues

Regardless of the optic technology used, images are created by combining light, which means the display has to compete with ambient light for visibility. This results in a significant loss of contrast, especially outdoors. Kerr et al. (2011), for example, report that participants repeatedly used their hands to shield the display from external light to be able to read its contents.

Many manufacturers of optical see-through displays equip them with shaded lenses to improve the contrast somewhat [Itoh et al., 2021]. Unfortunately, current OST technology is incapable of producing the contrast levels required for outdoor use [Erickson et al., 2020]. Technical solutions currently in development include OSTs that can partially occlude areas of the display to block light. However, these are still very experimental [Kiyokawa et al., 2000, Davari et al., 2020].

As a hardware-independent solution, Tanaka et al., 2008 propose a method for automatically placing content where readability is better, i.e. darker segments of the background. Such concepts might not be suitable for safety-critical scenarios, as users rely on spatially anchored elements to quickly acquire information.

3.1.2 Color Blending

In the additive color model, the combination of all wavelengths of light produce white, while black is created by the absence of light. When physical screens display black, the corresponding pixels do not emit light, so the actual color of the display material is visible. Since OHMDs do not have a physical screen, "black" is transparent [Gabbard et al., 2013]. This also means that darker colors are not available for design on OHMDs, as they are mostly translucent. Additionally, the level of transparency affects the extent to which the background distorts the color [Itoh et al., 2021].

Gabbard et al. (2013) additionally found that, depending on the background, colors may shift in hue or washout due to their luminance value and/or chromaticity. Both of these effects result in colors potentially being misinterpretated or not being visible at all.

4 The Human Visual System

4.1 The Field of View and Peripheral Vision

The human field of view spans approximately 200° to 220° horizontally and around 125° vertically, with the vertical field of view extending slightly further downward [Loschky et al., 2017]. The central vision covers about 5° around the point of fixation, where acuity is highest [Rosenholtz, 2016, Loschky et al., 2017]. The remaining field is the periphery, with boundaries at 30° and 60°, dividing it into near, mid and far peripheral vision.

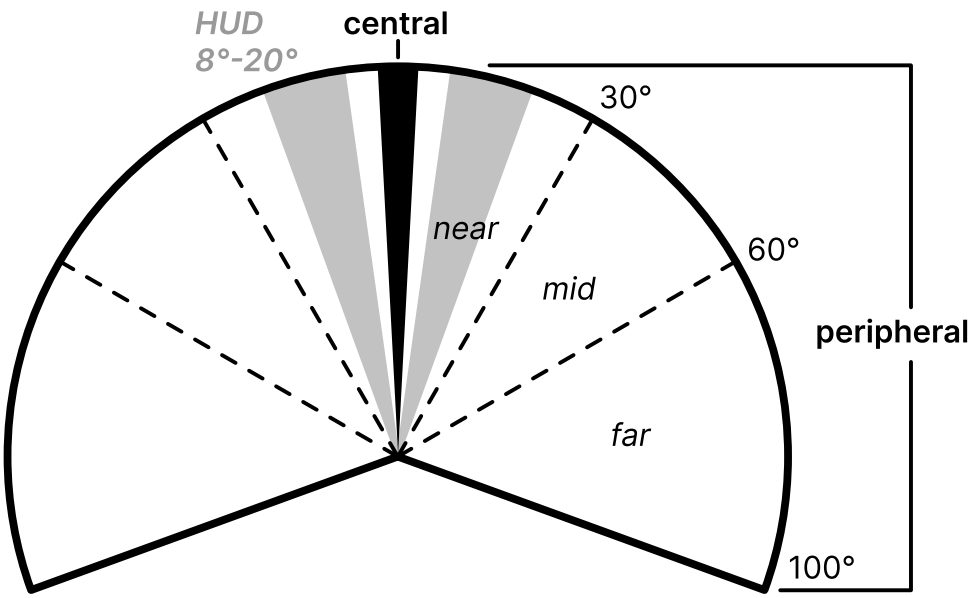

Figure 4: Simplified illustration of the horizontal field of view

Despite visual acuity and other functions deteriorating in the periphery, the area in front of us continues to appear in high resolution, with high contrast and color saturation [Loschky et al., 2017]. According to Rosenholtz (2016), the decline in performance of the peripheral vision is smaller than commonly assumed. A scene does not require many details to still allow object recognition and orientation, which is why image thumbnails remain recognizable despite their small size.

In addition, peripheral vision is particularly well suited for detecting motion, provides better vision in low-light conditions [Johnson, 2014] and contributes to functions like orientation, navigation, balance, and hand-eye coordination. A restriction of the peripheral visual field severely impairs the performance of these functions, the effect increasing with the amount of vision obstructed [Alfano and Michel,

1990].

Color sensitivity decreases toward the periphery, which is more pronounced for red-green colors than for blue-yellow colors and for brightness perception. Hansen et al. (2009) found that color perception is possible even at an eccentricity of 50°, provided the stimulus is sufficiently large. Most modern OST-HMDs feature a 40° field of view (see Section 3.1), meaning the entire available color spectrum can be utilized.

Rosenholtz (2016) suggests that crowding is the biggest issue in parsing information located in the periphery. When visual elements are placed too close together, they blend into a single entity, making it harder to distinguish them from one another. This effect is more pronounced if the target and its neighboring elements are similar (e.g., same color and shape) [Pelli and Tillman, 2008]. The farther elements are from the center of vision, the greater the distance between them must be.

4.2 Visual Attention

The retina transmits too much information to be processed all at once. This bottle neck is resolved by visual attention, which focuses resources on just a fragment of the current visual stimuli [Anderson et al., 2005, Rosenholtz, 2016].

Two processes determine where this attention is focused next. Bottom-up attention occurs involuntarily, triggered by external stimuli. In the top-down process, the gaze is directed toward objects related to current needs or goals. Interestingly, top-down attention reduces the impact of distracting stimuli while consciously focusing on a visual target [Connor et al., 2004].

An object is referred to as visually salient, if it attracts attention by being distinct from its surroundings in some way, without specifying which characteristics cause it to do so. When multiple stimuli compete for attention, even salient targets might be ignored if they are irrelevant to the current goal. This is referred to as inattentional blindness [Jensen et al., 2011].

5 Attention Management

After discussing the mechanisms of perception and attention, the question arises as to how they can be utilized and which aspects need to be taken into consideration in order to do so.

5.1 Situation Awareness

Making good judgements becomes increasingly difficult when performing demanding procedures while trying to maintain awareness of the surrounding area. The risks associated with these activities prohibit human error, further increasing stress levels. Endsley (1995) highlights the role of situation awareness as a foundation for good decision-making in dynamic situations. She outlines three levels of SA, namely perception, comprehension and projection, of which perception is the most significant for UI design. This stage requires collecting the information relevant to the situation. In level 2 this information will be analyzed to form "a holistic picture of the environment, comprehending the significance of objects and events." Finally, likely outcomes can be determined by understanding the dynamics of the situation.

It is evident that situation awareness is therefore related to the processes of attention described in the previous section. Subsequently, this illustrates the necessity of a thorough understanding of the pieces required for achieving good situation awareness to design a well-functioning HUD. The amount of information that can be rapidly and accurately obtained is directly influenced by how it is presented. In addition to prioritizing and organizing that information around task objectives, Endsley (1995) mentions the advantage of systems providing processed information rather than the data it is based on (e.g. displaying remaining time based on current oxygen amount and usage).

She emphasizes that special attention should be paid to reserve highly salient features to critical events that truly necessitate changing the course of action. Further, it is vital to maintain a balance between showing all relevant information, while simultaneously avoiding visual overload. While looking for inaccessible information, other important content may be overlooked. Conversely, "filtering of extraneous information (not related to SA needs) and reduction of data (by processing and integrating low-level data to arrive at SA requirements)" is necessary to prevent clutter. This reiterates the importance of accurately evaluating and prioritizing

task-relevant information.

As Endsley (1995) points out, considering these recommendations supports assessment of the interaction of all components as a whole, since emphasizing some elements inevitably comes at the expense of others. Evaluating the design concept by the amount of situation awareness it provides serves as the last step of ensuring the effectiveness of the system.

5.2 Peripheral Displays and Glanceability

The requirements of situation awareness make clear that information has to be quickly accessible, without straining cognitive capacity or being overly distracting. Peripheral displays do just that—keeping relevant information accessible while offering the choice of whether or not to continue with the current task [Matthews et al., 2005]. On the other hand, so-called notification displays actively interrupt the user to prompt necessary action.

Building onto this categorization, Matthews et al. (2006) introduces the concept of glanceability—a term used to describe information that is quickly and easily comprehended once it is paid attention to. Glanceability is directly impacted by the displayed amount of items, their salience, etc.—i.e. the attentional limits described in Section 4. In this context, Matthews et al. (2006) emphasizes that visual components need to "balance distinctiveness with too much demand for attention."

Lu et al. (2020) note that to design effective glanceable displays, it is necessary to consider both the visual presentation of information and how it is accessed. They present two novel concepts for hands-free interaction that implement the principles of glanceability and compare their performance to a regular HUD. Participants in the study were asked to follow a virtual mannequin at a specified distance for the primary task, while being asked questions about the information shown on the display.

The eye-glance interface corresponds to a conventional HUD, in which elements are fixed at the edges of the display's field of view. Although continuously displaying the information makes it easily accessible, it can lead to information overload or otherwise be distracting. Additionally, restricting the peripheral field of view can impair perception in that area.

The visual components of the head-glance interface are placed just outside the

display's field of view and are accessed by turning the head. Without this movement, the elements are not visible at all. This approach requires an additional sensor to track the head movement relative to the position of the body. According to Lu et al. (2020), the interface caused frustration among the participants. However, they theorize that this was due to the frequency of the questions asked, which led to numerous head movements within a short amount of time. In addition, participants found it mentally taxing to remember the location of specific information.

In the gaze-summon interface, visual targets are placed near the position where the interface elements would be located. If these points are fixated for a certain amount of time, the information appears at that location. However, in industrial environments this delay may not be justifiable in the intended applications. Additionally, Lu et al. (2020) report that keeping the gaze on target might lead to eye strain if the display is used frequently. There was also an increase of false positives when participants took a turn, caused by reflexively looking into the intended direction, combined with eye-tracking jitter from the rapid head movement.

These methods offer an interactive approach to countering information overload. That said, visual elements must also be designed in such a way as to draw the user's attention to the relevant information.

5.3 Capturing Attention

Peripheral displays are designed to enable passive monitoring of information. In critical situations, however, the display must be able to capture the user's attention, which can be achieved through the intentional use of motion.

In their "Peripheral Displays Toolkit" Matthews et al. (2004) demonstrate how to assign appropriate notification levels to information and what the transitions between these levels should look like. Abrupt transitions, for example, are meant to interrupt the user in order to attract their full attention. When aiming for divided attention, it should be sufficient to update smaller areas of the screen or to use slower animations.

Janaka, Zhao, and Chan (2023) implemented this approach and demonstrated that using fade-in animations reduces the impact of interruption. The effectiveness of the notifications depends on their duration, position, as well as the complexity of the main task.

6 Data Visualization

"One of the greatest benefits of data visualization is the sheer quantity of information that can be rapidly interpreted if it is presented well" [Ware, 2013].

Data visualization transforms data into information. Data itself is just noise, whereas information holds value—value that is created by the context through which the data is interpreted.

6.1 Guiding Visual Search

Information needs to be located before it can be interpreted. To make this happen as quickly as possible, certain design principles can be utilized.

To this end, Shao et al. (2023) present the 3V method, which refers to visual hierarchy, weight, and flow respectively. Visual hierarchy on the interface level uses unity, contrast, and order to sort and group information. The visual weight is described by the features of and space occupied by elements. It works through the distinction of objects, achieved by differing presentation. More visual weight means the object attracts more attention. The visual connections between elements describe the visual flow. If implemented correctly, flow guides the gaze to target information.

They evaluated this framework by testing visual search performance of four different label arrangements. The labels consisted of a string of numbers, and described boxes chaotically arranged on the screen. They found that visual search is more efficient, if the labels are arranged in vertical columns on the sides of the screen. This facilitates scanning of the content. Two reasons: if the labels are read left to right, it is easer to distinguish the start and end of them. Although the eye moves faster horizontally than vertically, which makes it easier to control jumps between rows.

In a similar vein, Matthews et al. (2006) explain: "Visuals that support top-down search will be easy to cognitively interpret. Visuals that support bottom-up search enable relevant information to pop-out."

In summary, even when a user is consciously (i.e., top-down) searching for a specific element in the user interface, a well-structured design can leverage bottom-up attention to guide this visual search [Shao et al., 2023].

6.2 Facilitating Rapid Comprehension

To make information glanceable, concepts of data visualization can be leveraged.

Wilke (2019) lists position, shape, size and color as basic elements for visualization. Position describes the placement of data, related to other points or axes of a diagram. As for color, hue and value should be distinguished, since they are suited for different kinds of data. Hues tend to work better for distinctive categories. Value for continuous data GRADIENTS, the eye is much more sensitive to differences in lightness (see Section 4). Hues can also be used for gradients (e.g. rainbow), but require much more intention to work.

Each of these elements can be used to encode information. Which element is used for which information influences the type of visualization. Importantly, this mapping must be unique, else it becomes ambiguous and obscures the meaning [Wilke, 2019].

matthewsToolkitManagingUser2004 "Abstraction involves extracting features or reducing the fidelity of information so that it is easier to read "at a glance" than the raw input. Abstraction enables lower attention consumption of information."

Information can be abstracted when the represented concept is simpler (e.g. ambient motion detector), in which case primitive elements of design, such as color and movement can be sufficient.

6.3 Visualizations for Single Value Data

Aside from design fundamentals, most specialized resources discuss visualization types intended for interrelated data. Interpreting such complex information requires time and cognitive capacity unavailable in scenarios such as those mentioned in Section 2. In these situations, a clear picture of the current status must be available at all times, which only allows for individual data values and therefore limits the number of possible visualization types.

Ferdio, an agency specialized on infographics and data visualization, has created a library of visualization types[3]. In this library, out of a 166 options, only nine are for single data (see Figure 5).

[3]Data Viz Project | Collection of data visualizations to get inspired and find the right type

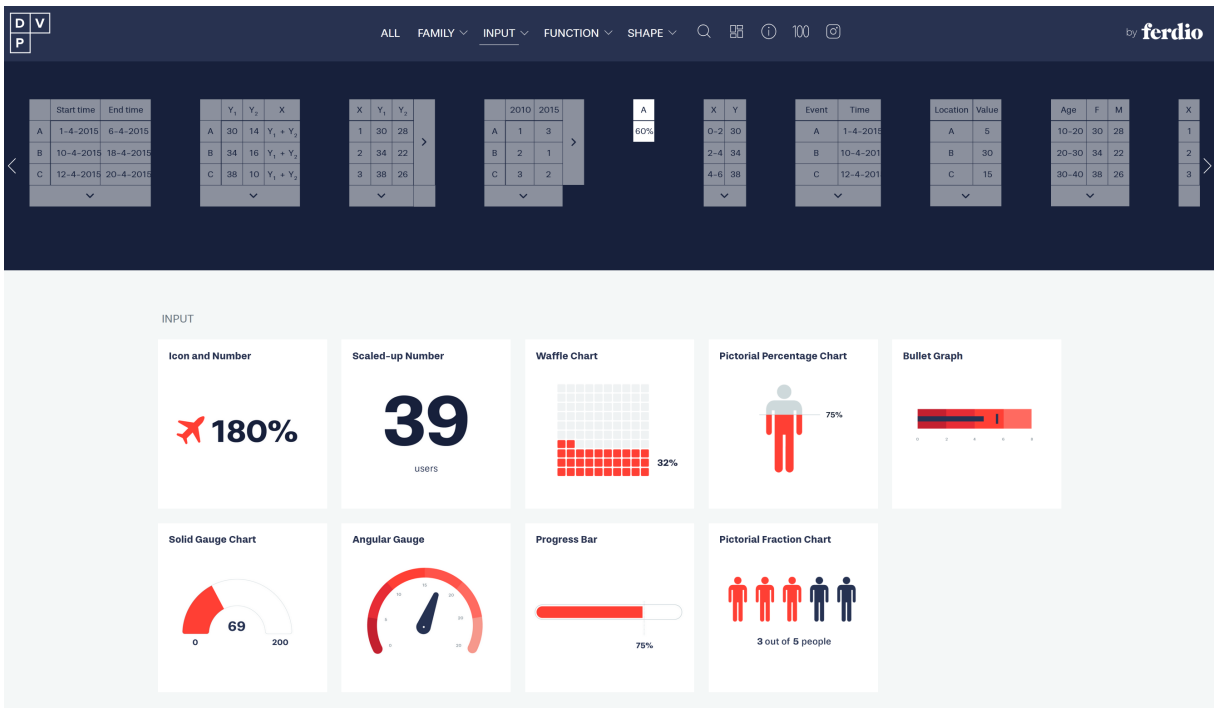

Figure 5: The options for single data (highlighted in the menu bar) in Ferdio's data visualization library.

7 Design Guidelines

The following guidelines are a compilation of the points discussed in the preceding sections, as well as recommendations from other authors. They serve as a basis for the design of head-up displays intended to support the efficient completion of objectives without compromising safety.

To facilitate situation awareness, the HUD should be capable of providing as much information as possible without causing fatigue or sensory overload. The displayed information should be comprehensible in the shortest possible time. In an emergency, the system must be able to capture the user's attention without hindering decision-making or completely interrupting the primary task.

7.1 Color and Contrast

As mentioned in Section 3, the use of colors in HMDs intended for daylight use is primarily limited by current optical see-through technology. Gabbard et al. (2022) recommend to prefer colors with high source luminance, as these are less prone to color shifts. Furthermore, if a color typically represents a specific meaning and cannot be guaranteed to be perceived correctly, alternative colors that can meaningfully convey that message should be considered. Red is specifically cited as an example, as it is highly susceptible to color shifts due to its low luminance.

The library chroma.js[4] has a function to sort colors by luminance in the RGB color space, based on the Web Content Accessibility Guidelines 2.0.[5] Sorting the base colors red (#FF0000), green (#00FF00), blue (#0000FF) and their intermediaries cyan (#00FFFF), magenta (#FF00FF), and yellow (#FFFF00) puts them into the following order (brightest to darkest): yellow, cyan, green, magenta, red, and blue.

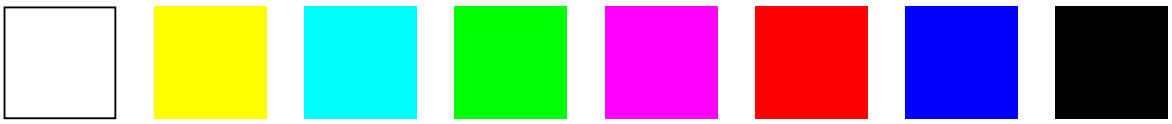

Figure 6: Colors sorted by highest to lowest luminance, in order: yellow, cyan, green, magenta, red, and blue. White and black have been added for reference.

[4] chroma.js api docs!
[5] Web Content Accessibility Guidelines (WCAG) 2.0

Limiting the UI to a single color can ensure that the design functions effectively in every respect (e.g., by using motion to draw attention rather than color). Not encoding information through colors also avoids potential semantic misinterpretations that may be caused by the previously mentioned color issues. In addition, monochrome displays are more affordable and therefore offer economic advantages, since replacement costs should be kept as low as possible in cases where equipment may be damaged or contaminated.

7.2 Typography

As with all head-mounted devices, walking causes them to shake, which affects legibility [Janaka, Zhao, and Chan, 2023]. Especially thin, horizontal lines tend to disappear optically due to the up- and down motion. Matsuura et al. (2019) confirm that heavier font weights improve readability. Naturally, the same requirements apply for stroke width in general.

For example, the typeface "Inter" meets these criteria well. It features a uniform strokes width, a square letter form and good line spacing. Compare this with the typeface "League Gothic". It also has a consistent stroke width, but due to its thickness, there is less white space within each character. Even with increased tracking, the narrowly shaped letters make reading considerably more difficult.

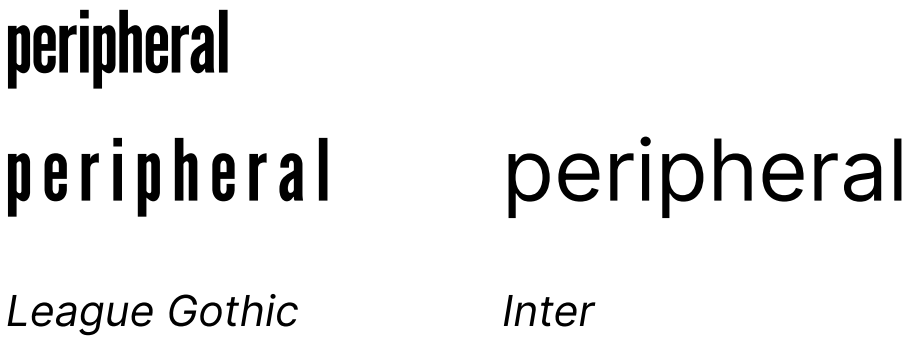

Figure 7: A comparison of the legibility of the typefaces "Inter" and "League Gothic," both set at a size of 32 px and in the regular weight. For "League Gothic," a line with increased character spacing was added, demonstrating that the font remains difficult to read.

To determine optimal font sizes, Renkewitz et al. (2008) tested to what extent the background and font colors affect text recognition. They set a benchmark of two seconds as threshold for fast recognition, which resulted in recommended font sizes from 23 to 42 pt. However, these values may have to be adjusted for use on

an OHMD, as they used a video see-through system and the colors red and blue, both of which have low luminance.

Gattullo et al. (2015) tested different text drawing styles in an indoor industrial setting. When the use of specific colors is mandated (e.g., in accordance with standard signage regulations), they recommend using white text on a billboard in the required color, which generally performed best. However, the billboard style obstructs much of the background. To remedy this, an outline style could be used, with thicker outlines further improving readability.

Figure 8: Billboard and outline text drawing styles

7.3 Iconography

Janaka, Zhao, and Sapkota (2023) tested if, and under which conditions icons perform better than text. They found that icon-augmented notifications enabled significantly higher primary task performance than text only and reduced the level of interruption while maintaining reaction and comprehension. They also found icons are more comprehensible when task complexity increased or demanded more attention. Additionally, the notification were "easier to perceive in suboptimal conditions", including outdoors when walking.

Naturally, this requires the icons to be easily recognizable. To that end, H. Lin et al. (2016) tested different versions of icons against each other. The differences tested were polarity, plane- and shape-based, and the presence or absence of borders. Since line-based icons dissect the space more frequently, they increase visual complexity, which impairs recognition performance. Overall, white shape-based icons on black background performed best.

Figure 9: Line-based and plane-based icons

7.4 Layout and Spacing

The ability of (binocular) head-mounted displays to display content anywhere within the field of view always presents a trade-off between immediate access to information and an unobstructed central line of sight. In use cases involving a great deal of physical activity, a clear field of view should of course be given greater priority.

Chua et al. (2016) investigated the positioning of content on a monocular OST-HMD worn in front of the right eye. The display was divided into a 3 $\times$ 3 grid, spanning 12.5° vertically and 17° horizontally in both directions from the center. The top and most peripheral positions were found to be more comfortable and unobtrusive, although middle and bottom center positions were perceived more quickly. The center-right is recommended as best balance between both usability and comfort. They mention that the ideal display position is highly dependent on task requirements (e.g., center is better when driving, secondary information better in peripheral).

It was found that the upper and most peripheral positions were perceived as more comfortable and less intrusive, although positions in the center and lower center were perceived more quickly. The position in the lower center is recommended as the best balance between usability and comfort. They note that the ideal display position depends heavily on the requirements of the specific task (e.g., the center is better for driving, while secondary information is better placed in the periphery).

G. Lin et al. (2021) divide HMD applications into four task categories: single (continuously focusing on display), alternating (switching back and forth between display and reality), background (passively monitoring the display for notifications) and dual (displayed information supplements physical primary task, e.g., captions). Similar to Chua et al. (2016), they mention that optimal positioning depends on task type. Participants preferred 5.4° to 14.6° horizontally, farther out was uncomfortable.

Ultimately, "there may be an ideal, general purpose HWD position between 8 to 20° temporally offset from line-of-sight. Horizontal FOV would thus be 12°."

Dowell et al. (2002) mention 8° around the line of sight should be kept free, since it "can induce cognitive tunneling and impair performance on the task over which the HUD is superimposed."

Clear zones to distinguish information

Crowding (see Section 4) affects perception the most. It seems advisable to prefer wider spacing between elements.

Additionally, the number of components per side should be limited to three at most, anchoring them spatially to the corners and sides of the display and keeping the center of vision free. **<empty citation>**

7.5 Notifications

- design according to required levels of attention and/or action

- encode each notification level uniquely

- use sparingly to avoid habituation and ensure the ability to capture the user's attention

The billboard style occludes much of reality around the text, which should generally be avoided. However, exactly this feature makes it suitable for high priority notifications.

Passive Monitoring	Inattention	Static
Active Monitoring	Divided Attention	Slow Animation, Partial Elements
Action Required	Focused Attention	Abrupt Animation

Table 1: Attention Levels and

8 Conceptualization

For printing, the color of the designs was inverted to black on white. A UI example in original colors is included as Figure 21.

8.1 Components

A prioritization of elements is necessary for decisions on visual hierarchy, as well as which ones can be left out if there is not enough space.

	Display Mode	Visualization
Oxygen	Continuous	Status Bar, Value
Hazard Proximity	Pop-Up	Icon
Warning Notification	Pop-Up	Billboard
Working Time	Pop-Up, Continuous	Countdown
Heart Rate	Continuous	Icon, Value, Trend
Ambient Temperature	Continuous	Icon, Value, Trend
Humidity	Continuous	Icon, Value
Checklist	Continuous	Line Highlight
Tool Help	Pop-Up	Icon, Value

Table 2: Parameters chosen for visualization, ordered from most to least important.

Continuous means that the component will be displayed permanently. Parameters marked with pop-up will be displayed occasionally when relevant.

8.1.1 Visualizing Oxygen Usage Through a Status Bar

Since the oxygen value is the crucial factor whether or not to continue work, this is the component taking the spotlight. As the most critical parameter, this component will be visible in any version of the HUD. To check the current status as quickly as possible, the value will be visualized with a status bar.

Since mechanical gauges with a pointer show visual progress by default, a radial version of the component was included. However, as mentioned in Section 6, linear design enable much quicker parsing of the information. Compared to the linear version, it also takes up much more space to make the information readable.

Figure 10: The initial concepts for the status bar, considering overall shape.

The second version places the value below the bar, on the right. When the amount of oxygen decreases, the empty portions of the bar are only a contour. Thus, the contrast between value and bar becomes stronger. The label is placed in front of the bar so they blend together, making it less salient. Additionally, the status bar was divided into quarter segments, enabling more accurate estimation of the amount left, without looking at the value.

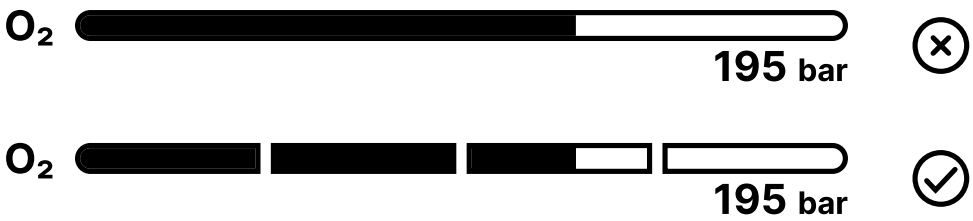

Figure 11: The modified status bar and the following segmentation.

Depending on available screen space and layout configuration, a vertical version of the component might be needed in addition to the horizontal one.

Grouping the label and value on one side saves space (left and middle of Figure 12). Aligning them at the top and bottom respectively accounts for visual crowding, making the value more distinctive. The oxygen value is located so that the status bar is seen first, i.e. if the component is placed on the left side, the bar will be on the right to maintain an optical guide.

Figure 12: Development of the vertical status bar.

8.1.2 The Base Component

This element is built with a simple configuration for secondary information that is helpful, but does not need to be prioritized. This includes data like heart rate, but also environmental information such as ambient temperature or humidity.

Values that are not to be visualized are similar enough to be based on a universal base component. Apart from the actual parameter value, any of these elements can be hidden. This allows flexible configuration depending on the type of information.

Figure 13: The base component with configurable icon, parameter value, unit and trend indicator.

The icons replace labels, and should therefore be as unambiguous as possible. Leaving the unit out saves space, though in some cases, it supports the identification of the value (e.g., degrees Fahrenheit or Celsius).

The trend indicators are displayed based on the difference of the current and previous value. The value increasing or decreasing an unusual amount might indicate events to take notice of. For dynamic elements like these, the frequency of changes should not be too high to avoid distraction. Comparing the short and long arrows, the long arrows are better readable when out of focus.

All of these values are displayed continuously, with the exception of the time component. It will only be displayed after a configured amount of time. The ele-

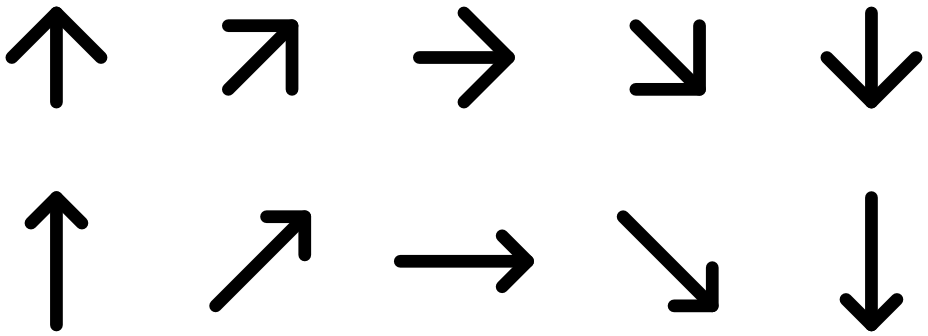

Figure 14: Comparison of short and long arrows.

ment popping up is meant to be perceivable in the peripheral vision, marking (for example) the halfway point, encoding this action with information. Since working time is limited by various parameters and therefore not fixed, the component is formatted as a countdown. To avoid pulling unnecessary attention, the display is limited to minutes, rather than ticking down by the second.

8.1.3 Notifications and Pop-Ups

Referencing typical industrial warning signs, the hazard icons were created in a triangular and a diamond-shaped version. Comparing them however, reveals that for the same size, the diamond shape appears smaller. This also means that there is less space for the graphic inside it. Avoiding unique silhouettes unites them in their function. For peripheral distinction, their placement is therefore more important. The icons are placed close together so that they are visible at the same angle if multiple hazards are present. Similar to a car's dashboard warning lights, each of them has a fixed position, facilitating rapid information acquisition. With regular notifications—depending on which one appears first—the order of content would be different.

The core example of the notification system is the high priority warning. This component is designed with a so-called billboard style, meaning text with a surrounding box. The big, bright area ensures high visibility even if the pop-up is not in the center of vision. Since it is the highest priority, it is the only element placed in the upper central screen area. To avoid blocking central vision for too long, the pop-up will transform into a smaller version located at [...].

Figure 15: Hazard indicator icons for electricity, explosive atmosphere and fire. The triangle and diamond shapes reference warning and hazard signage, respectively.

8.1.4 The Checklist Component

Since this element is text-based, it relies less on peripheral vision. To anchor the gaze to the current task, it is centered vertically and more distinctive than the other displayed tasks.

Note that this component is meant as a visual example, which is why certain aspects of functionality might not work in regard to handling the list. (e.g. if only two next tasks are displayed, there is no way to see if new tasks got added → task number display in dynamic situations, generally better for protocol).

The icon to the right is an anchor for possible gaze-based interaction, in this case to show more details. This design keeps the current task centered in the detailed view as well for faster orientation. This also mitigates side effects from possible false positives—i.e. if the system mistakenly opens the detailed view.

8.2 UI Examples

Note: the following illustrations are not to scale.

The maximum variant of the interface was designed with full functionality in mind, based on Microsoft's HoloLens 2. It features a horizontal field of view of 43°, a resolution of 1440 by 936 pixels per eye, and provides full color availability (within the technical limits mentioned in Section 3).

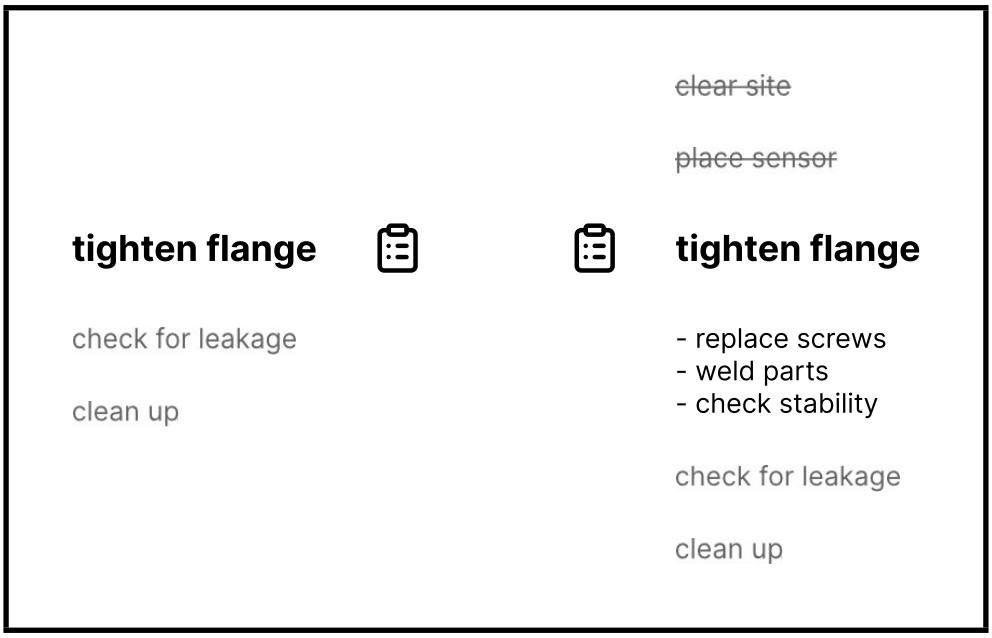

Figure 16: Default and expanded checklist. Note the center alignment of the current task to avoid losing focus when the list expands.

While the oxygen component is the most important, it is not necessarily the one that is checked most often; however, it should be the easiest to check. Therefore, it is placed in the lower center. Occasional information pop-ups are placed above the oxygen bar, for the same reason: looking down is faster than looking up.

The minimum version is based on the Google Glass, featuring a monocular display with a resolution of 640 by 360 pixels.

One could argue that some overlap of the components would be useful to make better use of the limited space. However, this would disrupt the grid pattern and the visual guides, resulting in longer search times.

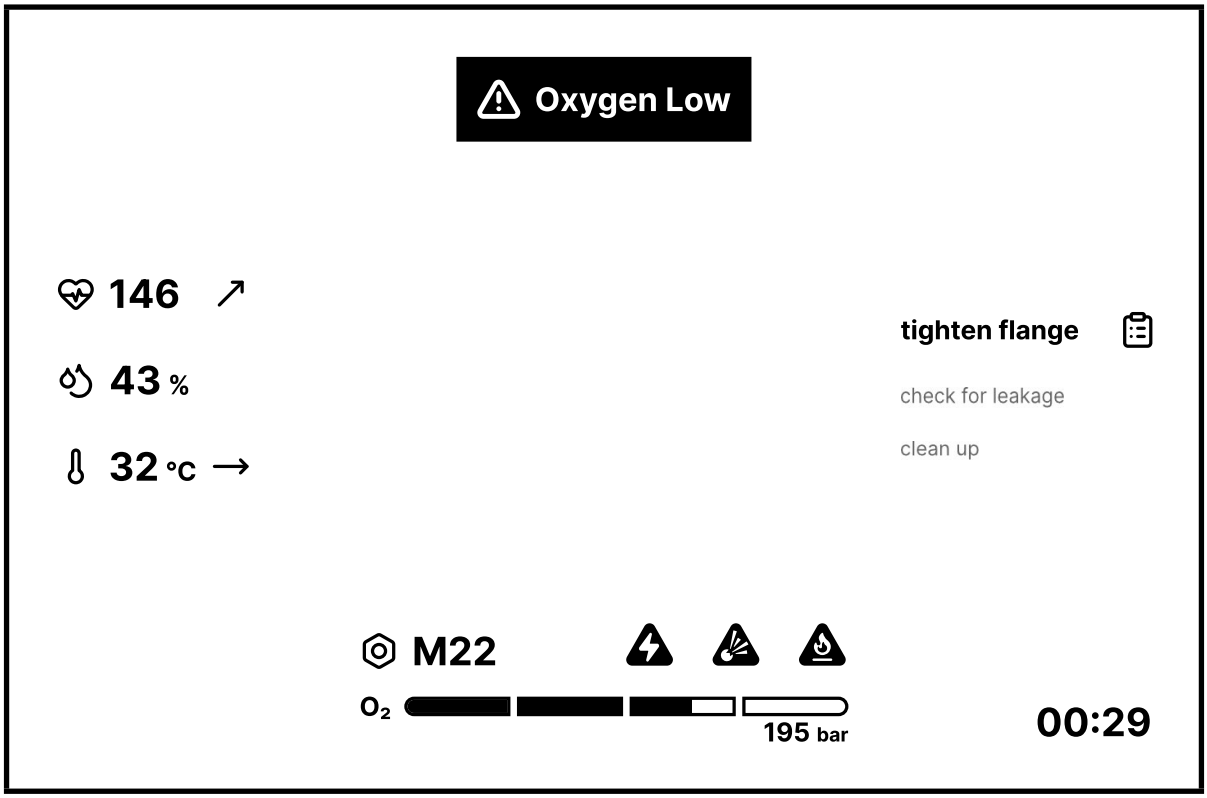

Figure 17: Full-featured UI concept for a binocular HUD.

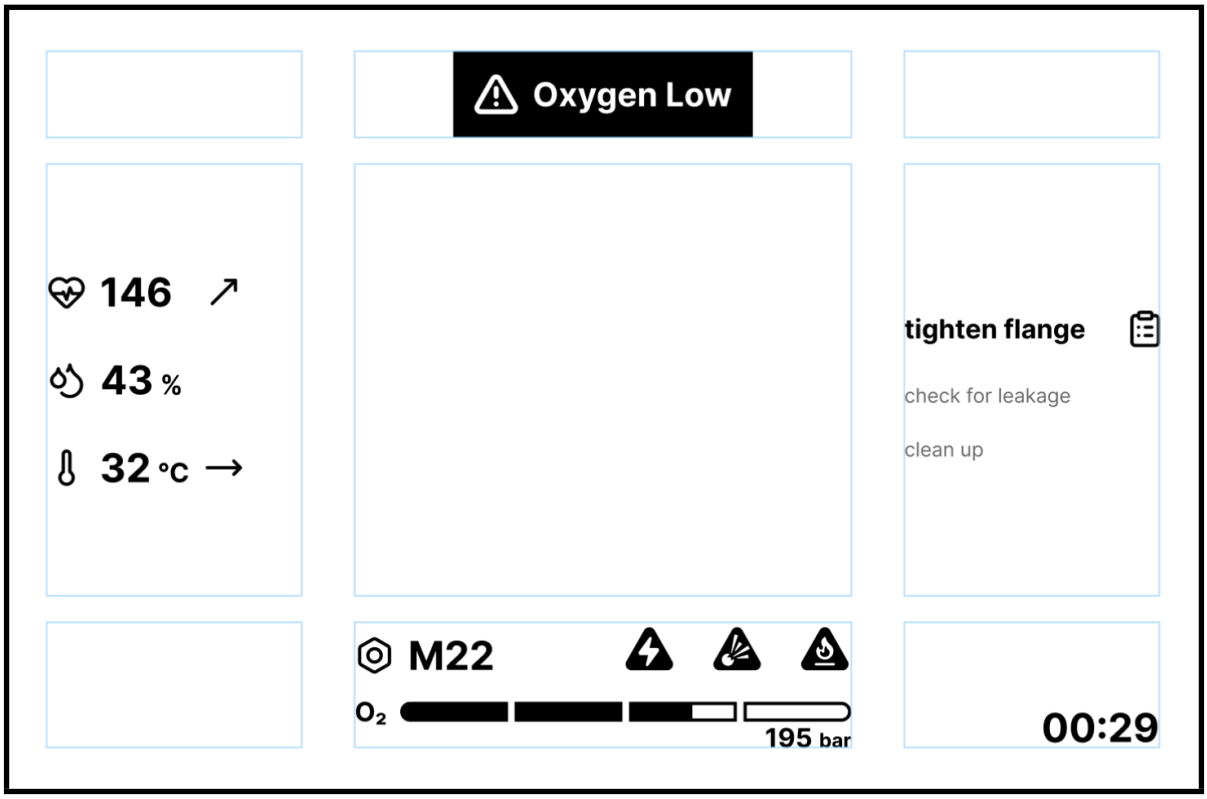

Figure 18: UI grid

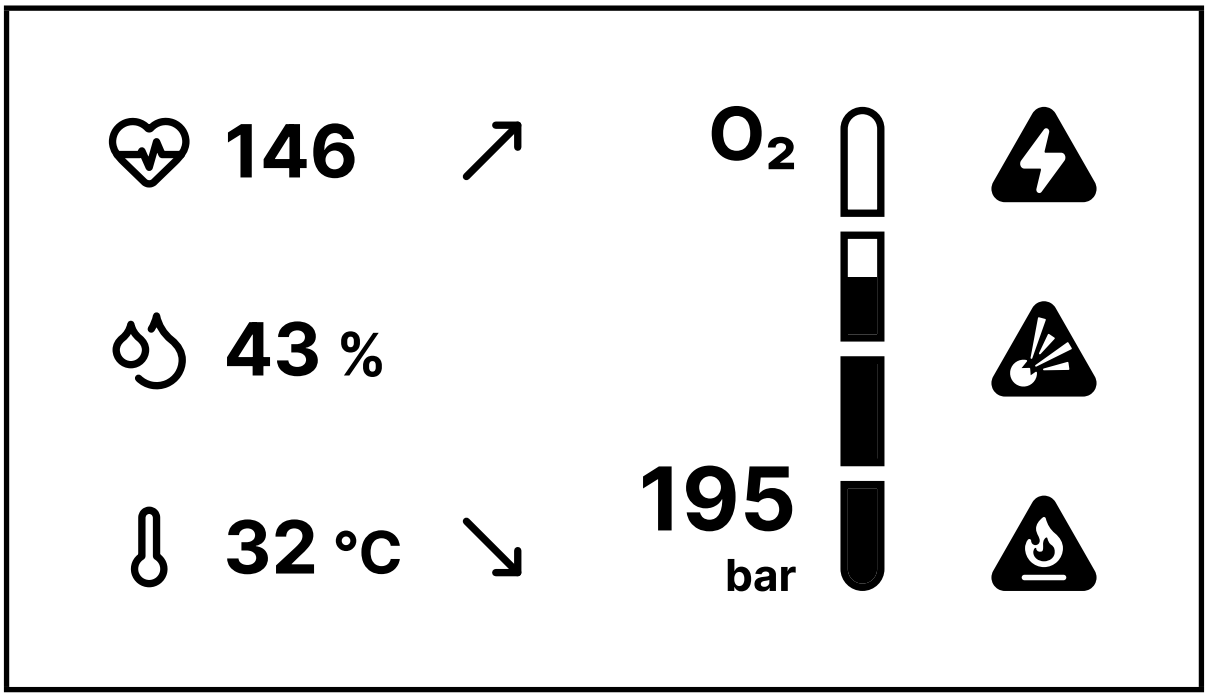

Figure 19: UI concept for a monocular HUD, worn in the upper left of the field of vision.

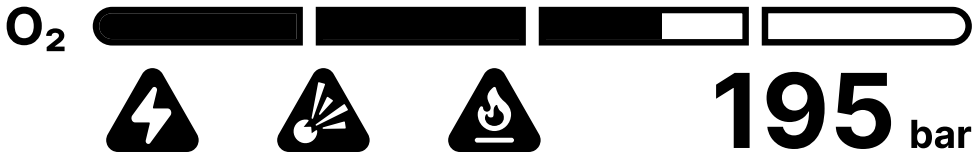

Figure 20: A negative example of overlapping elements. Placing the hazard symbols below the oxygen bar would save space, but disrupts the visual hierarchy.

Figure 21: HUD example

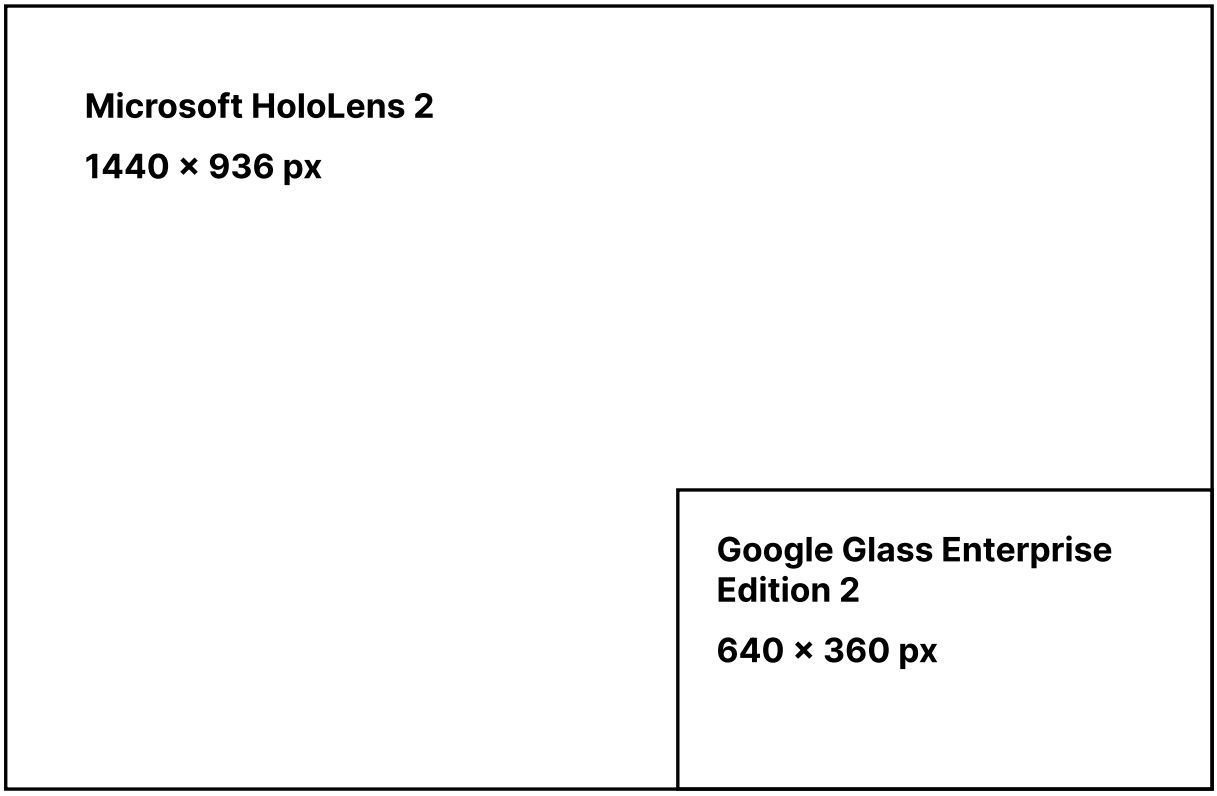

Figure 22: Size comparison of the Microsoft HoloLens 2 and the Google Glass Enterprise Edition 2

9 Conclusion

HUDs should be included in the AR spectrum as they are a tool for use cases that cannot be covered by world-stabilized AR.

Despite technological limitations, parts of system can already be created and tested testing and iteration: Hari Chandana et al., 2023

With the proposed building block system, different components for other use cases could be designed.

The implementation of a prototype was sadly outside the scope of this work. With this, animation could be tested.

Purely visual system for now, audio cues/multimodality: ZHU 2025

complete hands-free system

Appendix

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Hiermit erkläre ich, dass ich die vorgelegte Arbeit selbstständig verfasst, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie alle wörtlich oder sinngemäß übernommenen Stellen in der Arbeit als solche und durch Angabe der Quelle gekennzeichnet habe. Dies gilt auch für Zeichnungen, Skizzen, bildliche Darstellungen, Quellen aus dem Internet sowie für im Zusammenhang mit der Arbeit entstandene Produkte (z.B. Programmcode, Designentwürfe).

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Dresden, den 14.04.2026

Lloyd König